

Sneha Mishra

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LinkedIn | GitHub | LeetCode

SUMMARY

Software Engineering undergraduate with strong foundations in Data Structures Algorithms, Object-Oriented Programming, and Full-Stack Development. Experienced in building backend systems using Node.js, Express.js, MongoDB, and RESTful APIs, with working knowledge of React.js. Passionate about developing clean, scalable software and continuously learning modern software engineering practices.

EDUCATION

Indian Institute of Engineering Science and Technology, Shibpur Bachelor of Technology (Mechanical Engineering) CGPA: 8.26/10	Aug 2023 – Present
Indian Institute of Technology Madras Foundation Level, Data Science	May 2024 – May 2025

TECHNICAL SKILLS

- **Programming:** Java, C++, Python, JavaScript, TypeScript
- **Frontend:** React.js, Next.js, HTML, CSS
- **Backend:** Node.js, Express.js, REST APIs, JWT Authentication, MVC Architecture
- **Databases:** MongoDB, MySQL
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks, DBMS
- **Libraries:** NumPy, Pandas, Scikit-learn
- **Tools:** Git, GitHub, Postman

EXPERIENCE

Tata Steel – Research Intern	May 2026 – June 2026
• Worked under the guidance of the Head of TSCR Department on optimization of heat flux to reduce crack breakout in Thin Slab Casting and Rolling. • Utilized Python for data analysis and heat transfer evaluation to identify process parameters affecting casting quality. • Analyzed experimental datasets and generated insights that contributed to process optimization and reduced surface defects.	

PROJECTS

Hostel Complaint & Application Management System	Node.js, Express.js, MongoDB
• Developed a full-stack hostel management system with secure authentication and role-based authorization. • Designed RESTful APIs for complaint management, application processing, and administrative operations. • Implemented JWT authentication and MVC architecture to improve scalability and maintainability. • Built complaint lifecycle tracking with image uploads and real-time status updates. • Optimized MongoDB data models and backend APIs for efficient complaint retrieval and status management.	
Credit Card Fraud Detection Using Machine Learning	Python, Scikit-learn, Pandas, NumPy
• Built a machine learning model using Logistic Regression to detect fraudulent credit card transactions. • Performed data preprocessing, feature engineering, and handled class imbalance through undersampling. • Achieved 97.48% training accuracy and 98.32% testing accuracy. • Developed an automated prediction pipeline for real-time fraud classification. • Evaluated model performance using confusion matrix, precision, recall, and accuracy metrics.	

POSITIONS OF RESPONSIBILITY

Braking Lead, BAJA SAE Team – IEST Shibpur	May 2025 – Present
• Led the design and analysis of the braking system for the BAJA SAE vehicle, ensuring compliance with safety and performance standards. • Performed braking force, brake torque, and stopping distance calculations to optimize vehicle performance. • Collaborated with cross-functional teams during design reviews, testing, and validation of the braking subsystem.	

ACHIEVEMENTS

- **Flipkart GRID 7.0:** National Semi-Finalist.
- **Smart India Hackathon 2025:** College-Level Qualifier.
- **Reliance Foundation Scholar:** Awarded a national merit-based scholarship.
- **Diploma in Kathak:** Completed six years of training with First Division and Distinction.